Pranav Bhave

Cloud security × AI-assurance research. I build research software that exposes what an AI-assurance claim establishes, what it assumes, and where it must stop.

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In ninety seconds

Penn State CS '26 (Cybersecurity minor), AWS CCP + AI Practitioner. My research question: what do composed AI guardrails actually guarantee when their joint behavior was never measured? I maintain the Missing Column Census — twenty public guardrail evaluations in a bounded, single-reviewer inventory; five provide heterogeneous joint-evidence artifacts — four print a composition result and two release aligned per-item outcomes, with one artifact doing both — and the mathematics beneath it, CC-Framework, which answers with Fréchet–Hoeffding bounds and claim governance (the five-minute case study). At Penn State's S2 Lab I build Ghost-Ark, a verifier for the provenance limits of AI-governance receipts. Every claim on my site carries its evidence or says it can't — the evidence ledger has the bindings.

Research

AUG — DEC 2025 · PENNSYLVANIA STATE UNIVERSITY

Independent research — LLM safety guardrail composition (IST 496)

Supervised by Dr. Peng Liu. Reinforcement, interference, and dependence-driven failure in composed LLM safety systems using probabilistic bounds and copula-family reasoning. CC-Framework is the primary deliverable, with theorem notes, research memos, and architecture documentation.

JAN — MAY 2025 · PENNSYLVANIA STATE UNIVERSITY

Research assistant — logic & verification tooling

Python CNF-conversion tooling for SAT-solving and logical-verification workflows; evaluation of LLM-assisted formalization of natural-language logic — translation reliability, ambiguity handling, verification-readiness.

Selected projects

2026 · PYTHON, EVIDENCE ENGINEERING, PRIMARY-SOURCE RESEARCH

The Missing Column Census — a falsifiable field intervention

- A source-bound, bounded single-reviewer inventory of public guardrail evaluations: inclusion wording locked in repository history, 20 artifacts examined against primary sources, every row carrying passages, classification, and correction history. Corrected 2026-08-30 while retaining the 2026-08-27 cutoff: 14 document a shared item set and common event definition (the stricter ladder is 14 / 12 / 0); 5 provide heterogeneous joint-evidence artifacts — four print a composition result and two release aligned per-item outcomes, with one artifact doing both.
- The headline is recomputed from the census file in CI and registered as claim MC-001 with a fixed falsifier — the marketing claim is itself an executable, correctable object. Ships with a tested reference implementation of the proposed Minimum Joint Guardrail Disclosure.

2025 — PRESENT · PYTHON, STATISTICS, REPRODUCIBILITY

CC-Framework — dependence-aware AI assurance

- Treats composed-guardrail failure as partial identification: computes sharp Fréchet–Hoeffding bounds when rail marginals are known but joint dependence is not.
- Wraps results in claim envelopes, evidence roles, Merkle-logged receipts, and decay semantics; maintains an explicit claim-boundary manifest and non-claims.

IN DEVELOPMENT · S2 LAB, PENN STATE · AWS, TYPESCRIPT

Ghost-Ark — provenance limits of AI-governance receipts

- Verifier and measurement harness testing how far receipts identify executions — canonicalization kernels, collision corpora, and mechanically enforced non-claims.

EARLY STAGE · AWS NITRO ENCLAVES

Assay — enclave-attested media processing (private)

- Investigating which provenance claims a verifier can and cannot derive from Nitro Enclave attestation.

2026 · REACT/VITE · SECURITY ML

Ghost Visualizer · GCE · security-ML experiments

- Seven-scene visual essay on why isolated guardrail scores mislead; computes bounds, endpoint witnesses, and receipts from deterministic rows.

- Intrusion-detection and ML-backdoor evaluation experiments: attack-success-rate and false-negative-risk analysis.

Education & certifications

MAY 2026 · UNIVERSITY PARK, PA

Pennsylvania State University — B.S. Computer Science, minor in Cybersecurity

Coursework: computer security, operating systems, data structures, algorithms, statistical inference, theory of computation.

CERTIFIED · IN PROGRESS

AWS Certified Cloud Practitioner · AWS Certified AI Practitioner

SAA-C03 and Security — Specialty in progress. Verification IDs available on request.

Skills

CLOUD / AWS

IAM, S3, VPC, EC2, CloudWatch, Lambda, API Gateway, AWS Budgets, Bedrock-oriented architecture, Terraform basics, Nitro Enclaves (research)

AI / ML SECURITY

Guardrail composition, safety-stack evaluation, threat modeling, privacy auditing, intrusion detection, backdoor evaluation, ASR analysis

PROGRAMMING

Python, Java, C/C++, SQL, JavaScript/TypeScript; NumPy, Pandas, Scikit-learn, TensorFlow

PRACTICE

Reproducible experiments, claim governance, technical writing, Git, Linux, React, Node.js